Department of Computer Engineering

Lab Manual

310256 : Data Science And Big Data Analytics Laboratory

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Group B
Assignment No. 1

AIM: Write a code in JAVA for a simple Word Count application that counts the number of occurrences of each word in a given input set using the Hadoop Map-Reduce framework on local-standalone set-up.

COURSE OBJECTIVES: To gain practical, hands-on experience with statistics programming languages and Big Data tools

REQUIREMENTS:
Hadoop and MapReduce

THEORY:
In Hadoop, MapReduce is a computation that decomposes large manipulation jobs into individual tasks that can be executed in parallel across a cluster of servers. The results of tasks can be joined together to compute final results.

MapReduce consists of 2 steps:

- **Map Function –** It takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (Key-Value pair).

  Example – (Map function in Word Count)

Input	Set of data	Bus, Car, bus, car, train, car, bus, car, train, bus, TRAIN,BUS, buS, caR, CAR, car, BUS, TRAIN
Output	Convert into another set of data (Key,Value)	(Bus,1), (Car,1), (bus,1), (car,1), (train,1), (car,1), (bus,1), (car,1), (train,1), (bus,1), (TRAIN,1),(BUS,1), (buS,1), (caR,1), (CAR,1), (car,1), (BUS,1), (TRAIN,1)

- **Reduce Function –** Takes the output from Map as an input and combines those data tuples into a smaller set of tuples.

Example – (Reduce function in Word Count)

Input **(output of Map function)**	Set of Tuples	(Bus,1), (Car,1), (bus,1), (car,1), (train,1), (car,1), (bus,1), (car,1), (train,1), (bus,1), (TRAIN,1),(BUS,1), (buS,1), (caR,1), (CAR,1), (car,1), (BUS,1), (TRAIN,1)
Output	Converts into smaller set of tuples	(BUS,7), (CAR,7), (TRAIN,4)

Workflow of MapReduce consists of 5 steps:

1. **Splitting**
2. **Mapping**
3. **Intermediate splitting**
4. **Reduce**
5. **Combining**

Input:

Output:

Conclusion – Thus we created a word count program using Hadoop map-reduce.

Group B
Assignment 2

AIM: Design a distributed application using Map-Reduce which processes a log file of a system.

THEORY:

What is MapReduce?

MapReduce is a processing technique and a program model for distributed computing based on java. The MapReduce algorithm contains two important tasks, namely Map and Reduce. Map takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (key/value pairs). Secondly, reduce task, which takes the output from a map as an input and combines those data tuples into a smaller set of tuples. As the sequence of the name MapReduce implies, the reduce task is always performed after the map job.

1. **Steps for Compilation & Execution of Program:**

```
#sudo mkdir analyzelogs

ls

#sudo chmod -R 777 analyzelogs/

cd

ls

cd

..

pwd ls cd

pwd

#sudo chown -R hduser analyzelogs/

cd ls

#cd analyzelogs/

 ls cd ..
```

2. Copy the Files (Mapper.java,Reduce.java,Driver.java to Analyzelogs Folder)

#sudo cp /home/isha/Desktop/count_logged_users/* -/analyzelogs/

Start HADOOP

#start-dfs.sh

#start-yarn.sh

#jps

3. Compile Java Files

javac -d .

DsbdaMapper.java DsbdaReducer.java DsbdaDriver.java ls

#cd Dsbda/

ls cd .. #sudo gedit Manifest.txt

#jar -cfm analyzelogs.jar Manifest.txt Dsbda/*.class

ls cd jps

#cd analyzelogs/

4. Create Directory on Hadoop #sudo mkdir ~/input2000

ls pwd #sudo cp access_log_short.csv ~/input2000/

$HADOOP_HOME/bin/hdfs dfs -put ~/input2000 /

$HADOOP_HOME/bin/hadoop jar analyzelogs.jar /input2000 /output2000

$HADOOP_HOME/bin/hdfs dfs -cat /output2000/part-00000

stop-all.sh

jps

Input:

Output:

CONCLUSION: Thus, we have learnt how to design a distributed application using MapReduce and process a log file of a system

Group B
Assignment No. 3

AIM: Locate dataset (e.g., sample_weather.txt) for working on weather data which reads the text input files and finds average for temperature, dew point and wind speed.

THEORY:
Introduction to Map- Reduce:
MapReduce is a framework using which we can write applications to process huge amounts of data, in parallel, on large clusters of commodity hardware in a reliable manner. MapReduce is a processing technique and a program model for distributed computing based on java.
The major advantage of MapReduce is that it is easy to scale data processing over multiple computing nodes

The Algorithm
MapReduce program executes in three stages, namely:
1. Map stage
2. Shuffle stage
3. Reduce stage.

Inserting Data into HDFS:
•The MapReduce framework operates on pairs, that is, the framework views the input to the job as a set of pairs and produces a set of pairs as the output of the job, conceivably of different types.

• The key and the value classes should be in serialized manner by the framework and hence, need to implement the Writable interface. Additionally, the key classes have to implement the Writable -Comparable interface to facilitate sorting by the framework.

INPUT:

OUTPUT:

CONCLUSION: Thus, we have learnt how to process input data using map reduce framework.